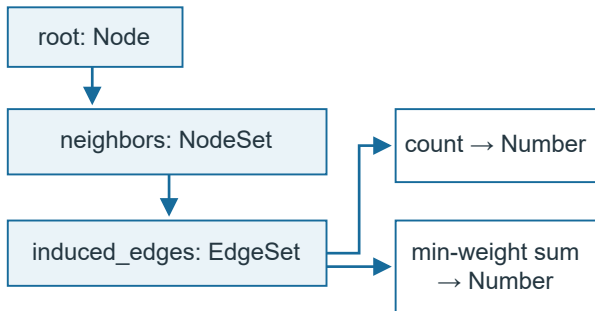
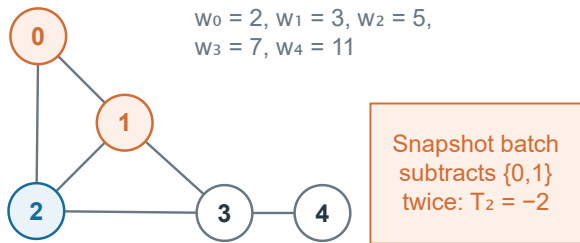


(a) Shared typed expression DAG



Share repeated subexpressions.
Cache only within the valid residual state.
Unrecognized expressions use the interpreter.

(b) Sequential residual deletion



Track surviving root $v = 2$

$T_2: 5 \text{ — delete } 0 \rightarrow 3 \text{ — delete } 1 \rightarrow 0$

Each step reads the updated residual neighborhood.

$T(v) \leftarrow T(v) - \sum \min(w_x, w_y), y \in N_R(v) \cap N_R(x), \text{ before deleting } x$

Preserve the interpreter's numeric interface; charge initialization, intersections, updates, caches and scoring.